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Anthony Katona, PhD

Summary Statement

Expert in polymer and ceramic coating processes, photothermal materials processing, and materials characterization. Experienced in lab-scale process development, class IV laser systems, safety oversight, and custom software and electronics development for increased productivity. Skilled at communication to a wide range of knowledge levels and building processes that improve transparency and maintainability.

Experience & Education

The Pennsylvania State University — *2023–current*
Chemistry (University Park, PA) — Postdoctoral Researcher

- led laser-processing development for refractory preceramic polymer coating precursors within an 8-group Multidisciplinary University Research Initiative, establishing experimental procedures, air-free workflows, and process-parameter screening methods for novel photothermal and photochemical ceramics pathways
- constructed remote monitoring instrumentation and a transparent flow/purge chamber for air-sensitive sample processing, introducing real-time tracking to existing laser treatment procedures
- designed nanosecond pyrometer to inform collaborating theory groups of accessible peak temperatures during laser pulses
- integrated containerization into multiple existing project software pipelines with complex dependency stacks to facilitate access to tools between collaborators

The Pennsylvania State University — *2023*
Chemistry (University Park, PA) — PhD

- developed photo-thermal polymer techniques for coatings and 3D-printing applications, determining mechanistic insights and novel methodology for spatiotemporal control of polymer properties
- modeled and printed custom 3DP components for managing optics and improving in-house HFR videography and contact angle goniometry
- synthesized ultra-fast/nano-scale thermal transport materials targeting space-limited, high-powered machines for military applications
- developed Python analysis pipelines for the conversion of raw profilometry, mechanical testing, DMA, and thermal instruments output to statistics and plots
- oversaw lab safety as both General and Laser Safety Officer, including management of chemical inventory and coordinating laboratory audits
- managed supply and maintenance of laboratory pumps, gas cylinders, gloveboxes, and various instruments
- taught as TA for all 5 first-year chemistry lecture and lab courses for a total of 8 semesters
- co-developed and implemented novel thermal conductivity undergraduate lab experiments for *Dr. Bratoljub Milosavljevic*'s physical chemistry course
- served as the direct graduate student mentor for 1/3 of entire incoming class of graduate students in the department and the research mentor for 5 undergraduate researchers, teaching experimental design, instrument use, scripting for data analysis, and scientific writing

Susquehanna University — *2015*
Biochemistry; Physics minor (Selinsgrove, PA) — BS

- synthesized transition metal Janus complexes with *Dr. William Doughtery* as a thesis project
- assisted in galaxy classification model training with *Dr. Violet Mager* during summer research

Technical Skills

wet lab: nanoparticle synthesis and dispersion, centrifugation, air-sensitive handling (Schlenk/glovebox), purification, crystallization, soxhlet extraction, electrochemistry, thermoset formulation/casting

spectroscopy/microscopy: NMR, IR, UV-Vis, AFM, TEM (STEM, EDS)

materials: mechanical (UTM, DMA), thermal (DSC, TGA, TPS), surface (profilometry, goniometry)

workshop: lathe, mill, 3D printing, laser engraver, hot press, CAD/CNC

laser: class IV CW/pulsed safety, handling, and optics

data/automation: Python (Pandas/NumPy/SciPy), C#, C++, LabVIEW, HTML, CSS, JS, SQL, ImageJ, WEKA segmentation, Docker, circuit design, Arduino/ESP32 microcontrollers,

technical communication: LaTeX, markdown, Adobe CC (Ae, Il, Ps, Pt, Pr), Veusz, FL Studio, Blender 3D

Publications & Patents

Publications:

- **A Katona**, S Phillips, J DiPietro, B Lear, (2026, submitted) Indust. Chem. & Mat., "Nanometer-Scale Surface Roughness of PDMS:CB Surfaces Compared to Traditional Cure Through Photothermal Parameterization"
- **A Katona** & B Lear, (2025). J. Phys. Chem. C, "Photothermal Curing of Polydimethyldisiloxane: Carbon Black Composites Results in Changes to Polymer Topography, Cross-Link Density, and Mass Density"
- A Blasone, **A Katona**, B Lear, (2025). RSC Appl. Poly. "Rapid photothermal curing of PDMS on paper"
- **A Katona** & B Lear, (2024). Macromol., "Effective Photothermal Curing of PDMS Using Ultralow Loadings of Carbon Black"
- V Mager, C Conselice, M Seibert, C Gusbar, **A Katona**, J Villari, B Madore, R Windhorst, (2018). ApJ 864, "Galaxy Structure in the Ultraviolet: The Dependence of Morphological Parameters on Rest-frame Wavelength"

Patent Filings:

- co-inventor, "**Photothermal Generation of Ceramics**," U.S. Provisional Patent Application No. 63/910,435, filed Nov. 3, 2025
- co-inventor, "**Photothermal Curing of Polymers on Thermally Sensitive Substrates**," U.S. patent application pending; priority applications filed July, November, and December 2024

Service & Outreach

Community Open Mic *2024-current*

- volunteer audio equipment, mixing, and recording services to weekly open-mic music events

Exploration-U *2019*

- co-presented polymer science demonstration at Bellefonte High School fair with >100 attendees

Graduate Women in Science *2017–2018*

- led demonstration on quantum locking in superconductors for 30 high school students
- designed and led a demonstration on the physics of sound for three classes of 20 Girl Scouts each
- designed and led a demonstration on the psychology of bias for three classes of 20 Girl Scouts each

Pennsylvania Junior Academy of Science *2017*

- served as a judge for student science presentations at the Penn State regional competition.